



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Textile Technology  
In-Semester Examination / Evaluation II  
Academic Year: 2025-26 (semester I)

Class: TY B. Tech.

Course: Man Made Fibre Technology

Course Code: PCC-TT305

Date: 10.10.2025

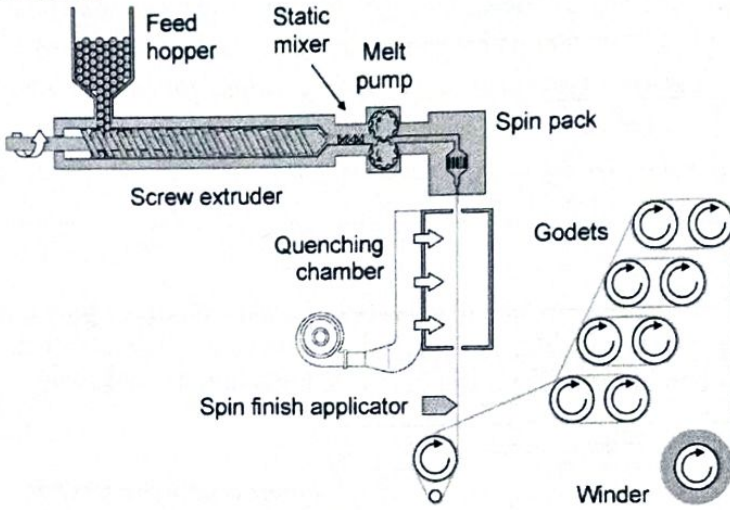
Time: 11:00 to 11:45

Max. Marks: 20

**Note:**

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

| Q.N. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks | CO  | BT  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| Q.1  | <p>A defence research unit has been tasked with developing a next-generation functional garment system for soldiers deployed on extended missions in extreme environments—ranging from humid jungles and arid deserts to high-altitude cold zones.</p> <p style="text-align: center;"><b>Functional Garment for Prolonged Military Missions (Extreme Environments)</b></p> <p style="text-align: center;">Figure: Illustration for a basic requirement for defence functional garment</p> |       |     |     |
| a    | Design a fibre blend using Nylon and PP for a military-grade functional garment. Justify the blend ratio and placement of each fibre based on moisture management, abrasion resistance, and thermal comfort.                                                                                                                                                                                                                                                                              | 7     | CO4 | BT4 |
| b    | List the mechanical and performance-related properties that should be evaluated for a functional garment (Do not list the properties already mentioned in the scenario).                                                                                                                                                                                                                                                                                                                  | 3     | CO3 | BT2 |

|                |                                                                                                                                                                                                                                                       |           |     |         |     |                |     |   |     |     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|---------|-----|----------------|-----|---|-----|-----|
| Q.2            | A Nylon 6 Technology has advantage over Nylon 66 and PET polymer.                                                                                                                                                                                     |           |     |         |     |                |     |   |     |     |
| a              | Discuss the key characteristics of Nylon 6 that make it suitable for continuous polymerization and fibre production.                                                                                                                                  | 3         | CO2 | BT3     |     |                |     |   |     |     |
| b              | Discuss the influence of different catalyst conditions on the polymerization of Nylon polymer with high molecular weight.                                                                                                                             | 3         | CO2 | BT3     |     |                |     |   |     |     |
| Q.3            | <p>Figure 2 Shows the schematic diagram of melt spinning machine used in the production of Nylon and PP filaments.</p>  <p style="text-align: center;">Figure2</p> |           |     |         |     |                |     |   |     |     |
| a              | <p>Match the following</p> <table><tr><td>Flat Yarn</td><td>BCF</td></tr><tr><td>Carpets</td><td>LOY</td></tr><tr><td>Fish net yarns</td><td>FDY</td></tr></table>                                                                                    | Flat Yarn | BCF | Carpets | LOY | Fish net yarns | FDY | 2 | CO3 | BT2 |
| Flat Yarn      | BCF                                                                                                                                                                                                                                                   |           |     |         |     |                |     |   |     |     |
| Carpets        | LOY                                                                                                                                                                                                                                                   |           |     |         |     |                |     |   |     |     |
| Fish net yarns | FDY                                                                                                                                                                                                                                                   |           |     |         |     |                |     |   |     |     |
| b              | Suggest the appropriate denier range and essential physical characteristics of polypropylene (PP) filaments suitable for developing upholstery fabrics.                                                                                               | 2         | CO3 | BT3     |     |                |     |   |     |     |

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

|      |                                                                      |
|------|----------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to: |
| CO 1 | Determine suitable method and parameters for spinning of a filament  |
| CO 2 | Determine process temperature looking to thermal behavior of a fiber |
| CO 3 | Compare properties of various fibers for probable application        |
| CO 4 | Select suitable spin finish for a fiber and process                  |
| CO 5 | Select high performance fiber for special applications               |
| CO6  | Determine thermal effect on fiber properties and structure           |



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.

**In-Semester Examination-II AY 2025-26**

SGGSJET, Nanded

Date: 09/10/2025

Class : Third Year Textile Technology

Time: 11.00 PM

Max. Marks : 20

Subject: PCC-TT302/303 Fabric Forming Technology- III

Duration: 45 Min

- Note: i. All questions are compulsory.  
ii. Assume suitable data if necessary

| Q.N. | Question                                                                                                         | Marks | CO | BT       |
|------|------------------------------------------------------------------------------------------------------------------|-------|----|----------|
| 1    | With a neat diagram, explain the basic principle of air-jet weaving.                                             | 5     | 2  | BT2, BT3 |
| 2    | Describe the role of relay nozzle and profile reed in air-jet looms. Why is it important for weft insertion?     | 3     | 2  | BT2, BT3 |
| 3    | Compare air-jet looms and projectile looms in terms of speed, efficiency, and fabric quality.                    | 2     | 4  | BT4      |
| 4    | a) List different projectile jaws involved in the filling insertion process.                                     | 2     | 1  | BT1      |
|      | b) Draw and label the torsion bar picking mechanism used in projectile looms.                                    | 3     | 2  | BT1, BT3 |
| 5    | a) What are the different types of projectiles used in projectile looms? Give their dimensions and applications. | 3     | 1  | BT1, BT2 |
|      | b) What is the function of projectile guide teeth in a projectile loom? Name its two types.                      | 2     | 1  | BT1, BT2 |

== ALL THE BEST ==

**Bloom's Taxonomy (BT):**

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

- CO 1: Understand conventional and new weaving technologies.  
CO 2: Write about features, principle motions of automatic and modern weaving systems  
CO 3: Apply knowledge of advance weaving systems in product manufacturing etc.  
CO 4: Compare the weaving technologies for appropriate selection.

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Textile Technology

In-Semester Examination / Evaluation II

Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Textile Testing-I

Course Code: PCC-TT301

Date: 11.10.2025

Time: 11:00 to 11:45

Max. Marks: 20

**Note:** All questions carry equal marks. Attempt any four questions.

| Q.N. |                                                                                                                                                            | Marks | CO  | BT  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| Q.1  | a) Define relative humidity.                                                                                                                               | 2     | CO1 | BT1 |
|      | b) Explain the factors affecting the regain in textiles.                                                                                                   | 3     | CO4 | BT2 |
| Q.2  | With the help of suitable figure explain the working of fibre fineness tester based on air-flow principle.                                                 | 5     | CO3 | BT2 |
| Q.3  | Discuss comb sorter analysis for fibre length measurement.                                                                                                 | 5     | CO1 | BT2 |
| Q.4  | a) Define random and periodic variations.                                                                                                                  | 2     | CO1 | BT1 |
|      | b) Explain the causes and their effect on yarn irregularity.                                                                                               | 3     | CO2 | BT2 |
| Q.5  | If the most uniform 30 tex polyester staple yarn has a CV of 18%,<br>a) What is the lowest CV one would expect for a 16 tex yarn produced from same fibre? | 5     | CO4 | BT5 |
|      | b) What would be the CV of a 2-ply yarn produced from 16 tex yarn?                                                                                         |       |     |     |

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

**Course Outcomes:**

|      |                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                                                                  |
| CO 1 | Understand the principle and method of measurement of various physical and mechanical properties of fibre, yarn and fabrics in the field of textiles. |
| CO 2 | Apply the knowledge of these measurement techniques for future applications.                                                                          |
| CO 3 | Analyse physical and mechanical properties of fibre, yarn and fabrics in the field of textiles.                                                       |
| CO 4 | Conduct experiments related various measurement and performance evaluation of the textile materials.                                                  |
| CO 5 | Propose and design innovative testing methods or modification for improved evaluation of textile materials in specific applications.                  |